



**SECP2523
DATABASE (WBL)
SECTION 02**

ASSIGNMENT 1

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QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	Order	Customer
Information	Iphone 13	Name

QUESTION 2

Answer the questions below based on the following relational schema:

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SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)
EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)
PROJECT (Proj_Code, Proj_Name, Proj_StartDate)
SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
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- a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	-
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	-
SCIENTIST_EXPERIMENT	-	Sc_ID Exp_NO

- b) Does the table exhibit entity integrity? Answer 'Yes' or 'No' and then explain your answer. (4M)

TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	Have a primary key
EXPERIMENT	Yes	Have a primary key
PROJECT	Yes	Have a primay key

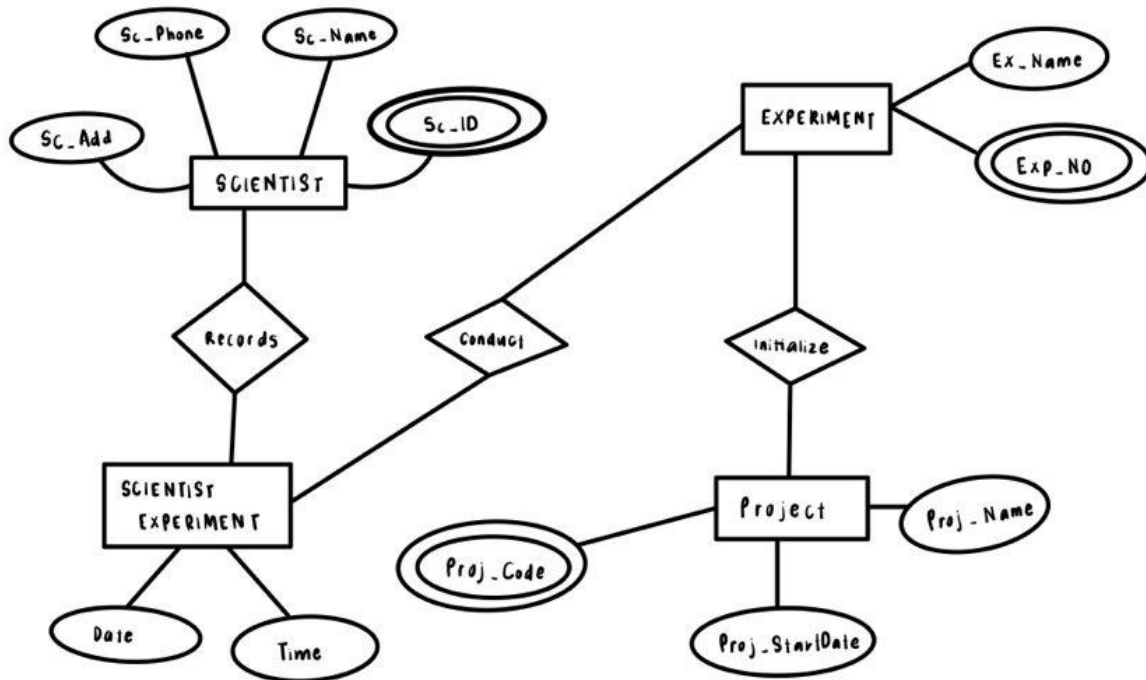
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SCIENTIST_EXPERIMENT	No	Does not have a primary key
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- c) Does the table exhibit referential integrity? Answer ‘Yes’ or ‘No’ and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY	EXPLANATION
SCIENTIST	No	does not reference another table. It only contains information about scientists and has its own primary key
EXPERIMENT	Yes	The EXPERIMENT table references the PROJECT table through Proj_Code. To maintain referential integrity, every value of Proj_Code in EXPERIMENT must exist in the PROJECT table.
PROJECT	No	The PROJECT table is not dependent on other tables; it only contains project information with Proj_Code as its primary key.
SCIENTIST_EXPERIMENT	Yes	The SCIENTIST_EXPERIMENT table references both the SCIENTIST table through Sc_ID and the EXPERIMENT table through Exp_NO. To maintain referential integrity, every value of Sc_ID in this table must exist in the SCIENTIST table, and every Exp_NO must exist in the EXPERIMENT table.

d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

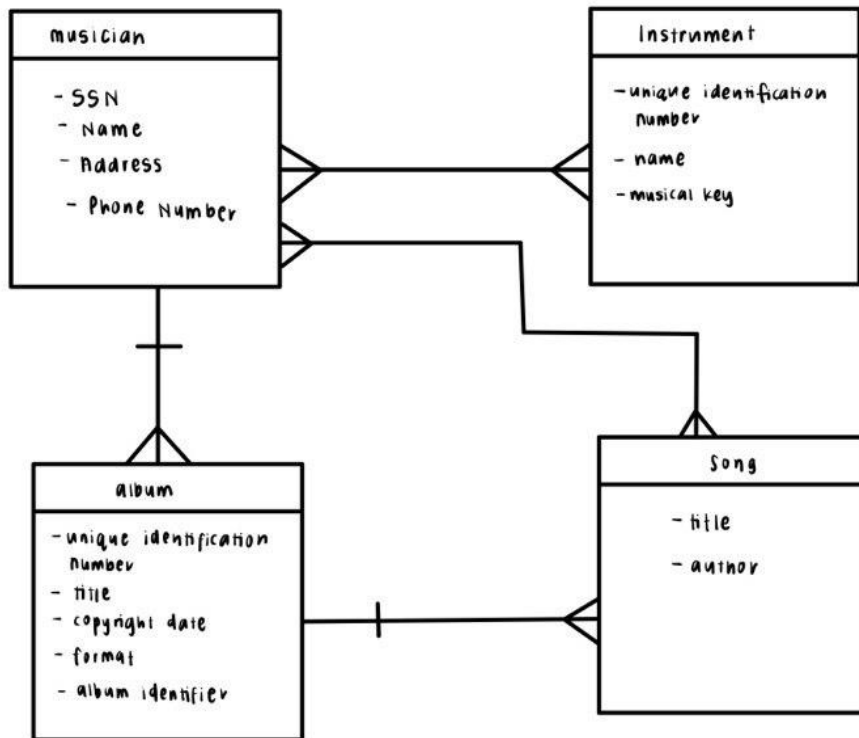
Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

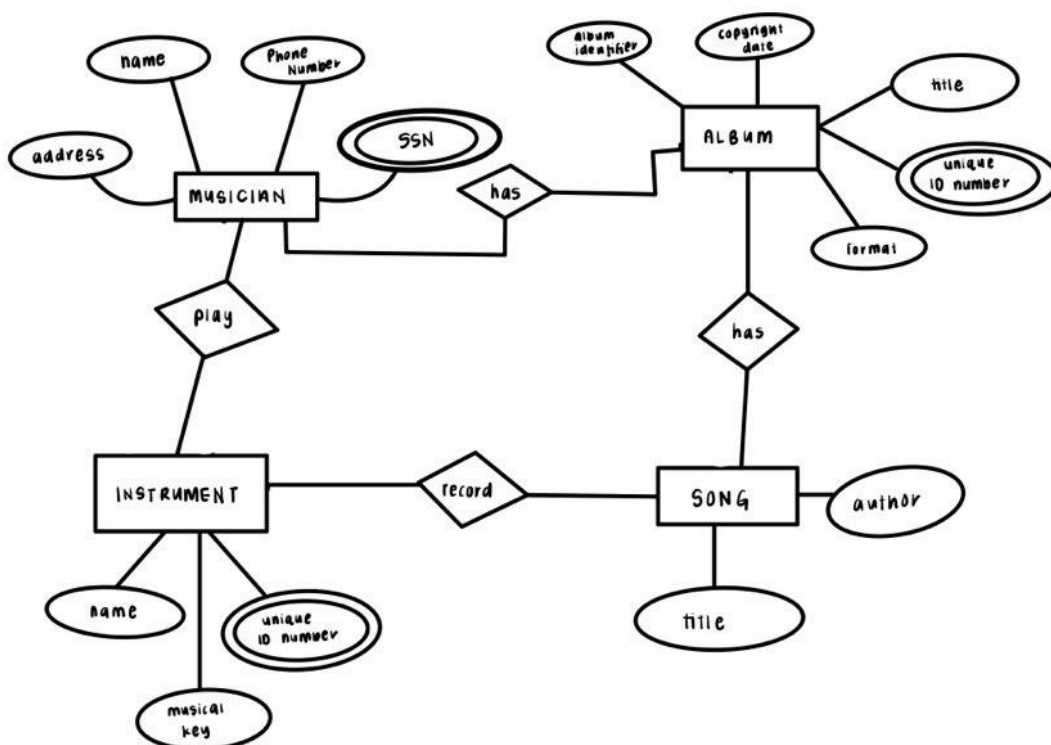
- Task:

1. List each Entity that is identified in Luxury Records System.
 - Musician
 - Instrument
 - Album
 - Song
2. For each entity list the attributes.
 - MUSICIAN (an SSN, a name, an address, a phone number)
 - INSTRUMENT (unique ID number, a name, a musical key)
 - ALBUM (unique ID number, a title, a copyright date, a format, an album identifier)
 - SONG (title, author)

3. Draw a logical model to represent each entity and the relationships between them.



4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.



QUESTION 4

Give an example of each of the three types of relationships.

- i) One-to-many relationship
A person has many social media accounts such as Instagram, X, TikTok and more
- ii) Many-to-many relationship
Food order placed by multiple customers and a customer may place more than one order
- iii) One-to-one relationship
A person that has his/her own car
- iv) Many-to-one relationship
Group of students need to accomplish their project in one month

QUESTION 5

CLIENT (Client_Number, Client_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)

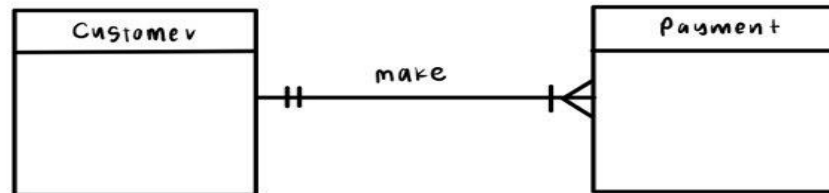
RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

- a) What is the relationship between Recruiter and Client?
Many-to-one relationships
- b) What is the primary key for the Client table and Recruiter table?
 - CLIENT (Client_Number)
 - RECRUITER (Recruiter_Number)
- c) Define foreign key? Identify foreign key in the Client table.
Foreign key is a column or combination of columns that is used to establish and enforce a link between the data in two tables to control the data that can be stored in the foreign key table.
Foreign Key: Recruiter_Number

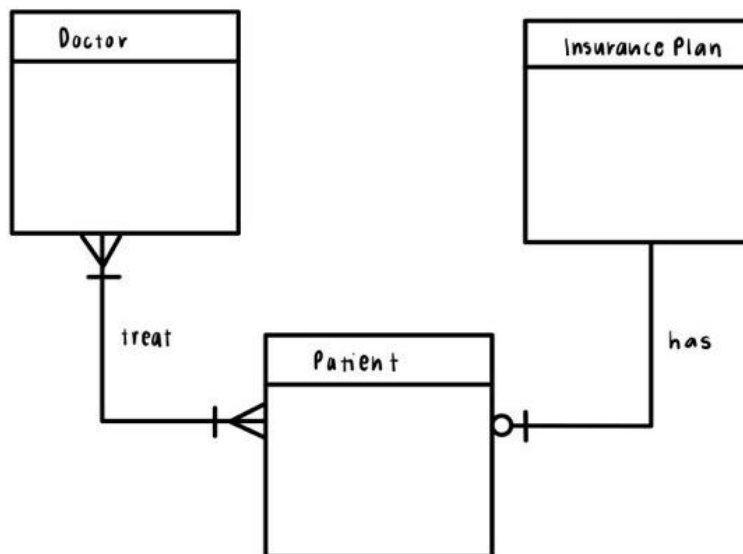
QUESTION 6

Draw the Chen & Crow's Foot notation based on the following statements:

- i) "A customer can make many payments, but each payment is made by only one customer."



- ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

STUDENT(SSN, Name, Major, Bdate)

COURSE(Course#, Cname, Dept)

ENROLL(SSN, Course#, Quarter, Grade)

BOOK_ADOPTION(Course#, Quarter, Book_ISBN)

TEXT(Book_ISBN, Book_Title, Publisher, Author)

Draw a relational schema diagram specifying the foreign keys for this schema.

